Controlled Class-Imbalance Construction for TCGA-UT

Yannik Queisler

June 22, 2026

Abstract

Class imbalance in computational pathology is routinely studied under real distributions, where prevalence, class separability, and data-collection history are entangled. This report defines a controlled TCGA-UT benchmark that varies only the long-tail severity of the training distribution while holding validation and test splits fixed. Power-law training distributions are generated under native-prevalence class ordering at imbalance parameters $\lambda \in \{0.8, 1.1, 1.3\}$ with a shared strictly feasible pool of 1,528 slides, yielding head-to-tail ratios of approximately 15:1, 43:1, and 78:1 that bracket the native TCGA-UT long tail. Using frozen Virchow2 patch and WSI-bag features with a 30-instance evidence budget, the same method families evaluated in a prior class-imbalance study on TCGA-UT (Queisler, 2026) are compared across three constructed severities. Method rankings are broadly preserved, while balanced accuracy declines monotonically by three to five percentage points from $\lambda = 0.8$ to $\lambda = 1.3$.

Contents

1 Introduction

A prior class-imbalance study on TCGA-UT compared mitigation methods under the dataset's native long-tailed slide distribution across 32 cancer types (Queisler, 2026). Native distributions are realistic but confound several factors simultaneously: how often each class appears, tumor morphology, tissue diversity, slide acquisition, and the representation geometry of the frozen encoder. A controlled construction isolates one factor—training-support frequency—by imposing known power-law distributions while holding validation and test data fixed.

TCGA-UT is a suitable test bed for this isolation because it provides slide labels and patch-level evidence from diagnostic whole-slide images in The Cancer Genome Atlas (Tomczak et al., 2015; Komura et al., 2022). Controlling only the long-tail severity under a fixed native-prevalence class ordering allows the question to be posed cleanly: do mitigation methods maintain their relative advantage as the imbalance between frequent and rare cancer types intensifies along the same ranking?

The report evaluates the same method families as Queisler (2026) under three constructed training distributions that vary only λ, the long-tail severity parameter, while holding the class ranking fixed. Patch-level methods include cross-entropy, class weighting, balanced sampling, focal loss, Scholz-style metric losses, CFAL, ProGAN augmentation, and OKO set learning (Scholz et al., 2024). WSI-bag methods include cross-entropy, class weighting, balanced sampling, focal loss, RankMix, SC-MIL, and MDE-MIL (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025). Patch and WSI-bag methods are reported separately because they consume different prediction units and answer regime-specific questions.

2 Dataset Construction

The construction shares the TCGA-UT case-disjoint split protocol with Queisler (2026). For each of three random seeds, the validation and test splits are held fixed, and artificial imbalance is applied only to the training split. The assignment is patient-disjoint: all slides and feature rows associated with one TCGA participant remain in exactly one partition. This isolates the effect of training-support frequency: model selection and evaluation are comparable across imbalance parameters and mitigation methods without patient leakage.

All 32 cancer types are retained. Let $C = 32$, let $\mathcal{D}_{\text{train}}$ denote the seed-specific training slide pool, and let a_c be the number of available training slides for class c. The constructed training split must satisfy

$$1 \leq n_c \leq a_c, \qquad \sum_{c=1}^{C} n_c = N, \qquad c \in \{1, \dots, C\}. \tag{1}$$

Every class receives at least one slide, no class is oversampled beyond its available pool, and the total training set is fixed at N slides shared across all imbalance parameters and seeds.

After slide selection, the input evidence is capped. Patch classifiers use up to 30 deterministic patches per slide. WSI-bag classifiers use up to 30 deterministic frozen feature instances per slide. In both regimes, Virchow2 features remain frozen (Vorontsov et al., 2024; Zimmermann et al., 2024), read from the shared `cls_patchmean` store (concat(CLS, mean(patch tokens)), 2,560-dimensional vectors). The cap aligns the input budget across regimes and prevents slides with more extracted tumor patches from dominating the training objective.

2.1 Power-Law Training Support

For an ordered class list $(c_1, \ldots, c_C)$, the intended share for rank i under imbalance parameter λ is

$$p_i(\lambda) = \frac{i^{-\lambda}}{\sum_{j=1}^{C} j^{-\lambda}}. \tag{2}$$

When $\lambda = 0$ the intended distribution is uniform over ranks; as λ increases, early-ranked classes receive more training support and late-ranked classes receive less. All constructed splits use the *native-prevalence order*: classes are ranked by their TCGA-UT training-split slide count, so rank 1 is the most frequent cancer type and rank $C = 32$ the least frequent.

Raw targets $N\,p_i(\lambda)$ are converted to integer counts by flooring and assigning residual slides by largest fractional remainder. Each class first receives one guaranteed slide; the remaining $N - C$ slides are then allocated according to Equation (2). A regime is accepted only when every rounded target satisfies Equation (1).

2.2 Feasible Pool Size and Realized Profiles

The pool size N must be small enough that no class target exceeds its available support at the steepest evaluated imbalance. Define the maximum feasible pool for a given λ and seed as

$$T^*(\lambda) = \min_c \frac{a_c}{p_c(\lambda)}. \tag{3}$$

For $N \le T^*(\lambda)$, all targets $N\,p_c(\lambda)$ are feasible. Setting $N = \lfloor \min_s T^*_s(1.3) \rfloor$ over the three seeds guarantees exact feasibility at the steepest evaluated imbalance on every seed. For the TCGA-UT training splits, this yields $N = 1{,}528$ slides, approximately 25% of the full available training pool.

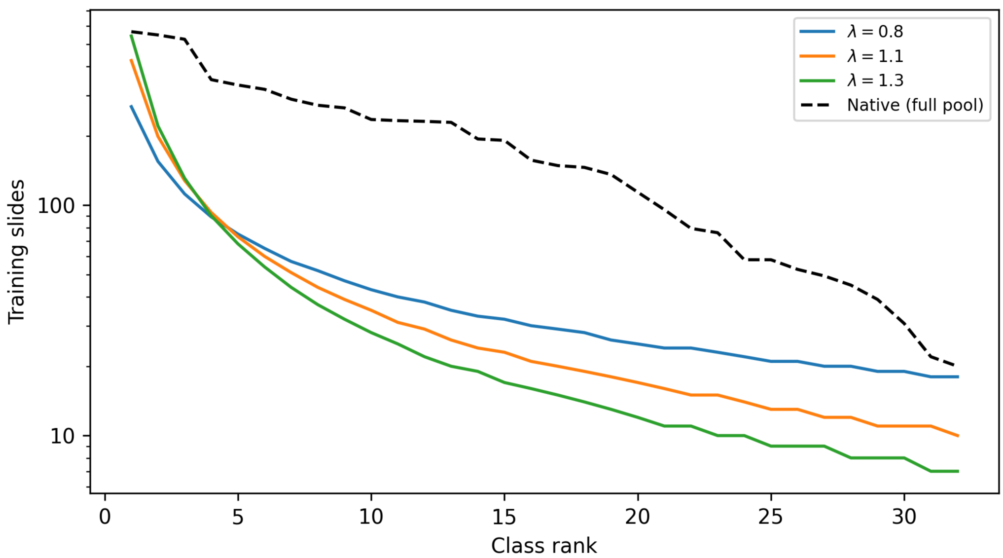

Figure 1: Constructed training support by class rank under the three evaluated imbalance parameters, with the native full-pool distribution shown as a reference. Classes are ordered by native prevalence (rank 1 = most frequent).

The three λ values produce head-to-tail ratios of approximately 15:1 ($\lambda = 0.8$), 43:1 ($\lambda = 1.1$), and 78:1 ($\lambda = 1.3$), bracketing the native TCGA-UT ratio of approximately 28:1 (Figure 1). The $\lambda = 0.8$ setting places the constructed distribution below native imbalance; $\lambda = 1.1$ is moderately more severe; and $\lambda = 1.3$ is the strongest regime evaluated.

3 Methods

The method taxonomy follows the companion benchmark, but this report uses it to study controlled training distributions rather than only the native TCGA-UT distribution. Cross-entropy is the no-mitigation reference. Data-level methods change minibatch exposure through balanced sampling or synthetic augmentation. Algorithm-level methods change the per-example loss through class weighting, focal loss, metric-derived objectives, prototype-affinity objectives, or OKO set supervision. WSI-specific methods change how slide bags are represented or mixed.

Family	Methods	Regime
No-mitigation baseline	CE / MIL CE	Patch, WSI bag
Data-level sampling and augmentation	Balanced sampling, RankMix	Patch, WSI bag
Algorithm-level losses	Weighted CE, Focal loss, CFAL	Patch (CFAL); Patch and WSI bag (others)
Hybrid sampling–loss	CE + soft F1, CE + soft MCC, OKO, MDE-MIL	Patch (first three, OKO); WSI bag (MDE-MIL)
Synthetic data generation	ProGAN augmentation	Patch
Representation and architecture	SC-MIL	WSI bag

Table 1: Method families evaluated under constructed TCGA-UT training distributions, following the taxonomy of Queisler (2026). Patch and WSI-bag methods are compared only within their own input regime.

Full method definitions follow Queisler (2026) and Scholz et al. (2024). RankMix, SC-MIL, and MDE-MIL are WSI-bag methods whose mechanisms operate on bags rather than isolated patch embeddings (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025).

4 Evaluation Protocol

Each constructed split is indexed by seed and imbalance parameter $\lambda \in \{0.8, 1.1, 1.3\}$. For every method and constructed regime, hyperparameters are selected by validation macro F1 across three seeds; ties are broken by validation balanced accuracy. Selected configurations are evaluated on the fixed test split, and results are reported as mean $\pm$ standard deviation across seeds. Macro F1 is the primary discrimination metric because it weights cancer types equally. Balanced accuracy and accuracy are reported as complementary discrimination metrics.

The comparison is intentionally regime-specific. Patch-feature methods are compared only with other patch-feature methods, and WSI-bag methods are compared only with other WSI-bag methods. The two regimes share the case-disjoint splits, frozen Virchow2 representation family, and 30-instance evidence budget, but they differ in prediction unit and aggregation mechanism.

Calibration is evaluated alongside discrimination using negative log likelihood (NLL), multiclass Brier score (Brier, 1950), and expected calibration error (ECE; Naeini et al., 2015; Guo et al., 2017). For each method and seed, a single temperature is fit by minimizing NLL on the validation split and applied unchanged to the test score matrix before softmax (Guo et al., 2017). Tables 5 and 6 report ECE before and after temperature scaling; full NLL and Brier metrics are given in Tables 7 and 8.

5 Results

The analysis asks how methods respond as imbalance severity increases from $\lambda = 0.8$ to $\lambda = 1.3$ under the native-prevalence class order. The realized support profiles are summarised in Table 2; Tables 3 and 4 compare mitigation methods within the patch-feature and WSI-bag regimes.

λ	Training slides	Support range	Head:tail ratio
0.8	1,528	18–268	≈15:1
1.1	1,528	10–425	≈43:1
1.3	1,528	7–543	≈78:1
native (full pool)	6,091	20–562	≈28:1

Table 2: Constructed training-support profiles under native-prevalence ordering. Support range and head:tail ratio are the minimum and maximum per-class slide counts averaged across seeds. The native full-pool row is shown for reference; it is not part of the constructed experiment.

Method	$\lambda = 0.8$		$\lambda = 1.1$		$\lambda = 1.3$	
	BAcc	F1	BAcc	F1	BAcc	F1
Weighted CE	0.714 ± 0.004	0.698 ± 0.006	0.698 ± 0.014	0.690 ± 0.016	0.679 ± 0.007	0.677 ± 0.013
Focal loss	0.718 ± 0.004	0.698 ± 0.004	0.696 ± 0.015	0.690 ± 0.018	0.672 ± 0.006	0.670 ± 0.013
CFAL	0.723 ± 0.004	0.699 ± 0.007	0.703 ± 0.013	0.705 ± 0.015	0.681 ± 0.010	0.685 ± 0.013
CE + soft F1	0.718 ± 0.005	0.685 ± 0.002	0.703 ± 0.020	0.680 ± 0.018	0.688 ± 0.007	0.668 ± 0.008
CE + soft MCC	0.709 ± 0.013	0.690 ± 0.007	0.692 ± 0.014	0.681 ± 0.020	0.668 ± 0.014	0.658 ± 0.012
Balanced sampling	0.712 ± 0.004	0.693 ± 0.006	0.694 ± 0.021	0.686 ± 0.027	0.672 ± 0.009	0.671 ± 0.009
OKO	0.723 ± 0.014	0.695 ± 0.020	0.707 ± 0.016	0.699 ± 0.018	0.683 ± 0.005	0.678 ± 0.003
ProGAN augmentation	0.723 ± 0.004	0.692 ± 0.010	0.698 ± 0.010	0.690 ± 0.011	0.677 ± 0.007	0.686 ± 0.010

Table 3: Patch-feature test results for validation-selected configurations under constructed training distributions.

Patch-feature results show a consistent, monotone decline as imbalance intensifies (Table 3). At mild imbalance ($\lambda = 0.8$), CFAL, OKO, and ProGAN augmentation lead in balanced accuracy (0.723), while CE+soft-F1 achieves the strongest performance at the steepest regime ($\lambda = 1.3$, 0.688); CE+soft-MCC is the weakest patch method throughout. Individual declines from $\lambda = 0.8$ to $\lambda = 1.3$ span three to five percentage points. Within the patch family the spread across methods at any fixed λ is approximately two percentage points—so the severity-induced decline is comparable to or larger than the advantage of choosing the best method.

Method	$\lambda = 0.8$		$\lambda = 1.1$		$\lambda = 1.3$	
	BAcc	F1	BAcc	F1	BAcc	F1
Weighted CE	0.802 ± 0.006	0.778 ± 0.003	0.794 ± 0.005	0.785 ± 0.001	0.789 ± 0.009	0.785 ± 0.012
Focal loss	0.794 ± 0.004	0.778 ± 0.006	0.800 ± 0.004	0.795 ± 0.008	0.786 ± 0.006	0.785 ± 0.003
Balanced sampling	0.809 ± 0.005	0.782 ± 0.004	0.791 ± 0.010	0.778 ± 0.020	0.777 ± 0.013	0.768 ± 0.017
RankMix	0.824 ± 0.018	0.799 ± 0.018	0.824 ± 0.017	0.804 ± 0.011	0.804 ± 0.008	0.790 ± 0.009
SC-MIL	0.812 ± 0.021	0.794 ± 0.015	0.801 ± 0.013	0.789 ± 0.015	0.782 ± 0.016	0.783 ± 0.011
MDE-MIL	0.824 ± 0.014	0.809 ± 0.006	0.811 ± 0.004	0.808 ± 0.007	0.804 ± 0.006	0.802 ± 0.016

Table 4: WSI-bag test results for validation-selected configurations under constructed training distributions.

The WSI-bag regime shows higher absolute performance than the patch regime (Table 4). MDE-MIL and RankMix lead at all three severities; RankMix is the most stable, declining by less than two percentage points from $\lambda = 0.8$ to $\lambda = 1.3$. Focal MIL shows a non-monotone response, while balanced-sampling MIL occupies the lowest tier at every severity. The ranking of bag-composition methods (RankMix, MDE-MIL, SC-MIL) over loss-based variants is maintained throughout, indicating that the advantage of WSI-specific aggregation mechanisms does not depend on imbalance severity.

Method	$\lambda = 0.8$		$\lambda = 1.1$		$\lambda = 1.3$		T
	ECE	ECE+TS	ECE	ECE+TS	ECE	ECE+TS	
Weighted CE	0.155 ± 0.009	0.012 ± 0.002	0.150 ± 0.010	0.009 ± 0.000	0.157 ± 0.017	0.013 ± 0.005	2.49 ± 0.03
Focal loss	0.113 ± 0.011	0.009 ± 0.002	0.110 ± 0.010	0.013 ± 0.002	0.122 ± 0.016	0.018 ± 0.003	1.78 ± 0.04
CFAL	0.738 ± 0.011	0.401 ± 0.009	0.734 ± 0.016	0.055 ± 0.006	0.720 ± 0.013	0.053 ± 0.005	0.05 ± 0.00
CE + soft F1	0.185 ± 0.007	0.015 ± 0.004	0.183 ± 0.012	0.018 ± 0.002	0.199 ± 0.014	0.016 ± 0.003	3.40 ± 0.27
CE + soft MCC	0.171 ± 0.005	0.011 ± 0.003	0.164 ± 0.010	0.017 ± 0.007	0.169 ± 0.006	0.022 ± 0.005	2.62 ± 0.12
Balanced sampling	0.156 ± 0.011	0.010 ± 0.002	0.157 ± 0.012	0.016 ± 0.006	0.166 ± 0.013	0.021 ± 0.001	2.43 ± 0.03
OKO	0.172 ± 0.007	0.039 ± 0.004	0.126 ± 0.017	0.035 ± 0.003	0.175 ± 0.017	0.033 ± 0.007	0.68 ± 0.03
ProGAN augmentation	0.195 ± 0.008	0.024 ± 0.004	0.194 ± 0.009	0.028 ± 0.005	0.206 ± 0.008	0.033 ± 0.001	4.24 ± 0.04

Table 5: Patch-feature calibration: ECE before (raw) and after (TS) temperature scaling across evaluated imbalance parameters. Fitted temperature T is averaged across seeds and severities. Full NLL and Brier metrics appear in Table 7.

Method	$\lambda = 0.8$		$\lambda = 1.1$		$\lambda = 1.3$		T
	ECE	ECE+TS	ECE	ECE+TS	ECE	ECE+TS	
Weighted CE	0.107 ± 0.007	0.023 ± 0.004	0.106 ± 0.003	0.022 ± 0.005	0.105 ± 0.012	0.018 ± 0.003	2.12 ± 0.06
Focal loss	0.080 ± 0.005	0.021 ± 0.002	0.052 ± 0.005	0.015 ± 0.000	0.060 ± 0.002	0.017 ± 0.001	1.44 ± 0.07
Balanced sampling	0.101 ± 0.006	0.019 ± 0.007	0.113 ± 0.003	0.020 ± 0.003	0.110 ± 0.012	0.022 ± 0.008	2.12 ± 0.06
RankMix	0.024 ± 0.007	0.017 ± 0.001	0.014 ± 0.004	0.018 ± 0.009	0.024 ± 0.001	0.021 ± 0.006	1.05 ± 0.01
SC-MIL	0.091 ± 0.009	0.018 ± 0.010	0.094 ± 0.010	0.019 ± 0.010	0.100 ± 0.010	0.022 ± 0.004	1.99 ± 0.02
MDE-MIL	0.042 ± 0.001	0.046 ± 0.012	0.054 ± 0.006	0.051 ± 0.006	0.091 ± 0.011	0.048 ± 0.007	0.79 ± 0.04

Table 6: WSI-bag calibration: ECE and fitted temperature T across evaluated imbalance parameters. Columns as in Table 5.

Probability Calibration. In the patch regime, CFAL is a calibration outlier: its affinity-based outputs are nearly flat over 32 classes, requiring strong temperature concentration ($T \approx 0.05$) and remaining substantially miscalibrated (ECE $\approx$ 0.40) at mild imbalance even after scaling. All other patch methods calibrate well with ECE below 0.04 after scaling, despite spanning a wide raw temperature range. In the WSI-bag regime, RankMix and MDE-MIL are already well calibrated before scaling; all methods reach ECE at or below 0.05 after temperature scaling. Calibration rankings are stable across imbalance severities, mirroring the discrimination pattern.

6 Discussion

The controlled construction reveals two complementary patterns. First, method rankings are broadly preserved across imbalance severities, with moderate reshuffling among closely performing methods. Second, balanced accuracy declines monotonically with λ for almost all methods, with a total range of three to five percentage points from the mildest to the strongest constructed regime. These patterns suggest that the relative utility of mitigation methods is largely robust to the particular severity chosen within this range, while absolute performance is sensitive to the support available for rare classes.

The monotone decline spans three percentage points for CE+soft-F1 to five for focal loss and ProGAN augmentation. The within-method spread at any fixed λ is approximately two percentage points—so the severity-induced decline is comparable to or larger than the advantage of choosing the best method. CE+soft-F1 shows the smallest decline and strongest relative performance at $\lambda = 1.3$, while CE+soft-MCC is the weakest patch method throughout.

In the WSI-bag regime, RankMix stands out for its stability: less than two percentage points of decline from $\lambda = 0.8$ to $\lambda = 1.3$, with performance unchanged from $\lambda = 0.8$ to $\lambda = 1.1$. Focal MIL is nearly flat across severities, showing a non-monotone response. MDE-MIL and SC-MIL decline somewhat more, while balanced-sampling MIL occupies the lowest tier at every severity. The ranking of bag-composition methods over loss-based variants is maintained throughout, indicating that the advantage of WSI-specific aggregation mechanisms is not contingent on the severity of training imbalance.

Calibration analysis complements discrimination: CFAL leads on patch discrimination but requires temperature scaling to produce reliable probability outputs, while MDE-MIL and RankMix improve both discrimination and raw calibration in the WSI-bag regime, though the gap relative to loss-based baselines narrows after a shared post-hoc scaling step.

Taken together, the controlled construction shows that when training-support frequency is imposed directly via a power-law distribution, aggregate discrimination declines predictably but modestly with severity, and the method ranking remains informative across the range tested. The prediction unit—patch versus WSI-bag—continues to be the dominant determinant of absolute performance: WSI-bag methods achieve balanced accuracy approximately ten percentage points higher than their patch-feature counterparts at equivalent imbalance.

7 Conclusion

This report defines a full-scale controlled TCGA-UT imbalance benchmark that keeps all 32 cancer types, preserves patient-disjoint validation and test splits, and varies only the training-support frequency under native-prevalence class ordering at $\lambda \in \{0.8, 1.1, 1.3\}$. A fixed pool of 1,528 slides yields head-to-tail ratios of approximately 15:1, 43:1, and 78:1, bracketing the native TCGA-UT long tail. Method rankings are broadly preserved across all three constructed severities, while balanced accuracy declines monotonically by three to five percentage points from the mildest to the strongest regime. At the patch level, leading methods at mild imbalance (CFAL, OKO, ProGAN augmentation) are partially reshuffled at the steepest severity, where CE+soft-F1 achieves the strongest performance; the within-method spread remains narrow enough that severity-induced decline is comparable to or larger than method-choice effects. At the WSI-bag level, bag-composition methods (RankMix, MDE-MIL, SC-MIL) consistently outperform loss-based variants regardless of severity, with RankMix showing the smallest decline. The prediction unit remains the dominant performance determinant: WSI-bag methods achieve balanced accuracy roughly ten percentage points above patch-feature methods at equivalent imbalance. Temperature scaling is sufficient to align probability outputs across all methods, though CFAL requires the largest correction in the patch regime.

References

Glenn W. Brier. Verification of forecasts expressed in terms of probabilities. *Monthly Weather Review*, 78(1):1–3, 1950. doi: 10.1175/1520-0493(1950)078⟨0001:VOFEIT⟩2.0.CO;2.

Yuan-Chih Chen and Chun-Shien Lu. RankMix: Data augmentation for weakly supervised learning of classifying whole slide images with diverse sizes and imbalanced categories. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23936–23945, 2023. doi: 10.1109/CVPR52729.2023.02397.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330. PMLR, 2017.

Dinkar Juyal, Siddhant Shingi, Syed Ashar Javed, Harshith Padigela, Chintan Shah, Anand Sampat, Archit Khosla, John Abel, and Amaro Taylor-Weiner. SC-MIL: Supervised contrastive multiple instance learning for imbalanced classification in pathology. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 7016–7025, 2024. doi: 10.1109/WACV57761.2024.00702.

Daisuke Komura, Akihiro Kawabe, Kenta Fukuta, Kazuya Sano, Tsukasa Umezaki, Haruka Koda, Riku Suzuki, Keisuke Tominaga, Michiaki Ochi, Hirofumi Konishi, et al. Universal encoding of pan-cancer histology by deep texture representations. *Cell Reports*, 38(9):110424, 2022. doi: 10.1016/j.celrep.2022.110424.

Xitong Ling, Yifeng Ping, Jiawen Li, Jing Peng, and Yuxuan Chen. MDE-MIL. *Dos mil tres mil*, 27, 2025. doi: 10.35707/dostresmil/27.

Mahdi Pakdaman Naeini, Gregory Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using bayesian binning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 29, pages 2901–2907, 2015. doi: 10.1609/aaai.v29i1.9602.

Yannik Queisler. Class imbalance mitigation for TCGA-UT. Technical report, TU Berlin, 2026.

Daniel Scholz, Ayhan Can Erdur, Josef Buchner, Jan C. Peeken, Daniel Rueckert, and Benedikt Wiestler. Imbalance-aware loss functions improve medical image classification. In *Proceedings of the 7th International Conference on Medical Imaging with Deep Learning*, volume 250 of *Proceedings of Machine Learning Research*, pages 1341–1356. PMLR, 2024.

Katarzyna Tomczak, Patrycja Czerwinska, and Maciej Wiznerowicz. The cancer genome atlas (TCGA): an immeasurable source of knowledge. *Contemporary Oncology*, 19(1A):A68–A77, 2015. doi: 10.5114/wo.2014.47136.

Eugene Vorontsov, Alican Bozkurt, Adam Casson, George Shaikovski, Michal Zelechowski, Kristen Severson, Eric Zimmermann, et al. A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine*, 30(10):2924–2935, 2024. doi: 10.1038/s41591-024-03141-0.

Eric Zimmermann, Eugene Vorontsov, Julian Viret, Adam Casson, Michal Zelechowski, and George Shaikovski. Virchow2. *2024 Shell TechXplorer Digest*, 2024(1), 2024. doi: 10.52196/208316.24.1.

A Full Calibration Metrics

Method	λ	NLL	NLL+TS	Brier	Brier+TS	ECE	ECE+TS	T
Balanced sampling	$\lambda = 0.8$	1.339 ± 0.059	0.841 ± 0.023	0.386 ± 0.019	0.335 ± 0.013	0.156 ± 0.011	0.010 ± 0.002	2.441 ± 0.095
Balanced sampling	$\lambda = 1.1$	1.371 ± 0.068	0.869 ± 0.038	0.386 ± 0.029	0.338 ± 0.022	0.157 ± 0.012	0.016 ± 0.006	2.389 ± 0.140
Balanced sampling	$\lambda = 1.3$	1.456 ± 0.084	0.913 ± 0.037	0.411 ± 0.022	0.354 ± 0.013	0.166 ± 0.013	0.021 ± 0.001	2.467 ± 0.000
CE + soft F1	$\lambda = 0.8$	1.868 ± 0.051	0.877 ± 0.009	0.414 ± 0.014	0.336 ± 0.009	0.185 ± 0.007	0.015 ± 0.004	3.334 ± 0.195
CE + soft F1	$\lambda = 1.1$	1.783 ± 0.051	0.906 ± 0.027	0.419 ± 0.025	0.349 ± 0.016	0.183 ± 0.012	0.018 ± 0.002	3.116 ± 0.104
CE + soft F1	$\lambda = 1.3$	2.150 ± 0.184	0.933 ± 0.052	0.439 ± 0.030	0.352 ± 0.021	0.199 ± 0.014	0.016 ± 0.003	3.763 ± 0.072
CE + soft MCC	$\lambda = 0.8$	1.524 ± 0.070	0.856 ± 0.036	0.404 ± 0.015	0.340 ± 0.012	0.171 ± 0.005	0.011 ± 0.003	2.789 ± 0.108
CE + soft MCC	$\lambda = 1.1$	1.487 ± 0.068	0.888 ± 0.041	0.397 ± 0.023	0.344 ± 0.020	0.164 ± 0.010	0.017 ± 0.007	2.581 ± 0.132
CE + soft MCC	$\lambda = 1.3$	1.490 ± 0.054	0.946 ± 0.041	0.421 ± 0.016	0.367 ± 0.013	0.169 ± 0.006	0.022 ± 0.005	2.495 ± 0.048
CFAL	$\lambda = 0.8$	3.338 ± 0.002	1.394 ± 0.032	0.960 ± 0.000	0.506 ± 0.011	0.738 ± 0.011	0.401 ± 0.009	0.050 ± 0.000
CFAL	$\lambda = 1.1$	3.212 ± 0.010	0.944 ± 0.063	0.950 ± 0.001	0.333 ± 0.022	0.734 ± 0.016	0.055 ± 0.006	0.050 ± 0.000
CFAL	$\lambda = 1.3$	3.219 ± 0.002	0.993 ± 0.047	0.951 ± 0.000	0.348 ± 0.014	0.720 ± 0.013	0.053 ± 0.005	0.050 ± 0.000
Focal loss	$\lambda = 0.8$	0.977 ± 0.040	0.793 ± 0.021	0.350 ± 0.015	0.323 ± 0.011	0.113 ± 0.011	0.009 ± 0.002	1.749 ± 0.068
Focal loss	$\lambda = 1.1$	0.984 ± 0.032	0.805 ± 0.033	0.348 ± 0.023	0.323 ± 0.019	0.110 ± 0.010	0.013 ± 0.002	1.750 ± 0.088
Focal loss	$\lambda = 1.3$	1.084 ± 0.081	0.865 ± 0.028	0.373 ± 0.017	0.343 ± 0.010	0.122 ± 0.016	0.018 ± 0.003	1.831 ± 0.122
OKO	$\lambda = 0.8$	1.038 ± 0.066	0.885 ± 0.066	0.369 ± 0.024	0.332 ± 0.024	0.172 ± 0.007	0.039 ± 0.004	0.651 ± 0.022
OKO	$\lambda = 1.1$	0.968 ± 0.039	0.872 ± 0.041	0.350 ± 0.019	0.328 ± 0.022	0.126 ± 0.017	0.035 ± 0.003	0.711 ± 0.028
OKO	$\lambda = 1.3$	1.080 ± 0.032	0.922 ± 0.054	0.379 ± 0.008	0.341 ± 0.015	0.175 ± 0.017	0.033 ± 0.007	0.665 ± 0.013
ProGAN augmentation	$\lambda = 0.8$	2.341 ± 0.128	0.862 ± 0.032	0.422 ± 0.016	0.334 ± 0.012	0.195 ± 0.008	0.024 ± 0.004	4.205 ± 0.000
ProGAN augmentation	$\lambda = 1.1$	2.448 ± 0.142	0.879 ± 0.051	0.417 ± 0.021	0.331 ± 0.014	0.194 ± 0.009	0.028 ± 0.005	4.205 ± 0.000
ProGAN augmentation	$\lambda = 1.3$	2.556 ± 0.155	0.917 ± 0.038	0.443 ± 0.017	0.348 ± 0.013	0.206 ± 0.008	0.033 ± 0.001	4.300 ± 0.082
Weighted CE	$\lambda = 0.8$	1.311 ± 0.079	0.799 ± 0.030	0.376 ± 0.017	0.321 ± 0.014	0.155 ± 0.009	0.012 ± 0.002	2.524 ± 0.098
Weighted CE	$\lambda = 1.1$	1.282 ± 0.048	0.809 ± 0.034	0.374 ± 0.025	0.322 ± 0.019	0.150 ± 0.010	0.009 ± 0.000	2.468 ± 0.082
Weighted CE	$\lambda = 1.3$	1.323 ± 0.137	0.839 ± 0.047	0.389 ± 0.025	0.333 ± 0.016	0.157 ± 0.017	0.013 ± 0.005	2.468 ± 0.082

Table 7: Patch-feature calibration: NLL, Brier score, and ECE before (raw) and after (TS) temperature scaling, with fitted temperature T. One row per method per imbalance parameter.

Method	λ	NLL	NLL+TS	Brier	Brier+TS	ECE	ECE+TS	T
MDE-MIL	$\lambda = 0.8$	0.614 ± 0.055	0.594 ± 0.059	0.223 ± 0.019	0.220 ± 0.020	0.042 ± 0.001	0.046 ± 0.012	0.831 ± 0.016
MDE-MIL	$\lambda = 1.1$	0.651 ± 0.021	0.623 ± 0.024	0.233 ± 0.007	0.228 ± 0.008	0.054 ± 0.006	0.051 ± 0.006	0.813 ± 0.032
MDE-MIL	$\lambda = 1.3$	0.714 ± 0.042	0.652 ± 0.059	0.251 ± 0.014	0.238 ± 0.016	0.091 ± 0.011	0.048 ± 0.007	0.735 ± 0.014
Balanced sampling	$\lambda = 0.8$	0.874 ± 0.041	0.599 ± 0.016	0.260 ± 0.011	0.237 ± 0.008	0.101 ± 0.006	0.019 ± 0.007	2.066 ± 0.040
Balanced sampling	$\lambda = 1.1$	1.002 ± 0.033	0.651 ± 0.036	0.283 ± 0.013	0.252 ± 0.011	0.113 ± 0.003	0.020 ± 0.003	2.208 ± 0.042
Balanced sampling	$\lambda = 1.3$	0.973 ± 0.102	0.690 ± 0.099	0.299 ± 0.042	0.269 ± 0.038	0.110 ± 0.012	0.022 ± 0.008	2.091 ± 0.119
Focal loss	$\lambda = 0.8$	0.670 ± 0.015	0.571 ± 0.009	0.247 ± 0.005	0.233 ± 0.005	0.080 ± 0.005	0.021 ± 0.002	1.548 ± 0.052
Focal loss	$\lambda = 1.1$	0.628 ± 0.040	0.574 ± 0.035	0.237 ± 0.015	0.231 ± 0.013	0.052 ± 0.005	0.015 ± 0.000	1.401 ± 0.047
Focal loss	$\lambda = 1.3$	0.643 ± 0.047	0.591 ± 0.037	0.248 ± 0.015	0.240 ± 0.014	0.060 ± 0.002	0.017 ± 0.001	1.385 ± 0.026
Weighted CE	$\lambda = 0.8$	0.859 ± 0.039	0.575 ± 0.013	0.258 ± 0.010	0.231 ± 0.004	0.107 ± 0.007	0.023 ± 0.004	2.137 ± 0.108
Weighted CE	$\lambda = 1.1$	0.897 ± 0.042	0.599 ± 0.035	0.266 ± 0.013	0.238 ± 0.013	0.106 ± 0.003	0.022 ± 0.005	2.184 ± 0.042
Weighted CE	$\lambda = 1.3$	0.841 ± 0.090	0.593 ± 0.047	0.269 ± 0.025	0.240 ± 0.020	0.105 ± 0.012	0.018 ± 0.003	2.043 ± 0.040
RankMix	$\lambda = 0.8$	0.552 ± 0.011	0.549 ± 0.010	0.224 ± 0.003	0.224 ± 0.002	0.024 ± 0.007	0.017 ± 0.001	1.062 ± 0.054
RankMix	$\lambda = 1.1$	0.574 ± 0.052	0.572 ± 0.052	0.223 ± 0.025	0.224 ± 0.024	0.014 ± 0.004	0.018 ± 0.009	1.049 ± 0.020
RankMix	$\lambda = 1.3$	0.584 ± 0.040	0.584 ± 0.040	0.235 ± 0.020	0.235 ± 0.020	0.024 ± 0.001	0.021 ± 0.006	1.050 ± 0.054
SC-MIL	$\lambda = 0.8$	0.755 ± 0.114	0.550 ± 0.053	0.245 ± 0.019	0.224 ± 0.018	0.091 ± 0.009	0.018 ± 0.010	1.977 ± 0.101
SC-MIL	$\lambda = 1.1$	0.790 ± 0.093	0.569 ± 0.046	0.252 ± 0.022	0.230 ± 0.021	0.094 ± 0.010	0.019 ± 0.010	1.979 ± 0.135
SC-MIL	$\lambda = 1.3$	0.792 ± 0.085	0.577 ± 0.046	0.263 ± 0.019	0.237 ± 0.015	0.100 ± 0.010	0.022 ± 0.004	2.021 ± 0.067

Table 8: WSI-bag calibration: NLL, Brier score, and ECE before and after temperature scaling. Columns as in Table 7.